

External Memory

By : Bilal Ahmed

External Memory



Also known as **secondary storage**.



Used to **store data permanently**.



Non-volatile — retains data even when power is off.



Provides **large storage capacity** at **low cost**.



Slower than main (primary) memory.



Data must be **loaded into main memory** before CPU can process it.

Common examples:



Hard Disk Drives (HDDs)



Solid-State Drives (SSDs)



USB Flash Drives



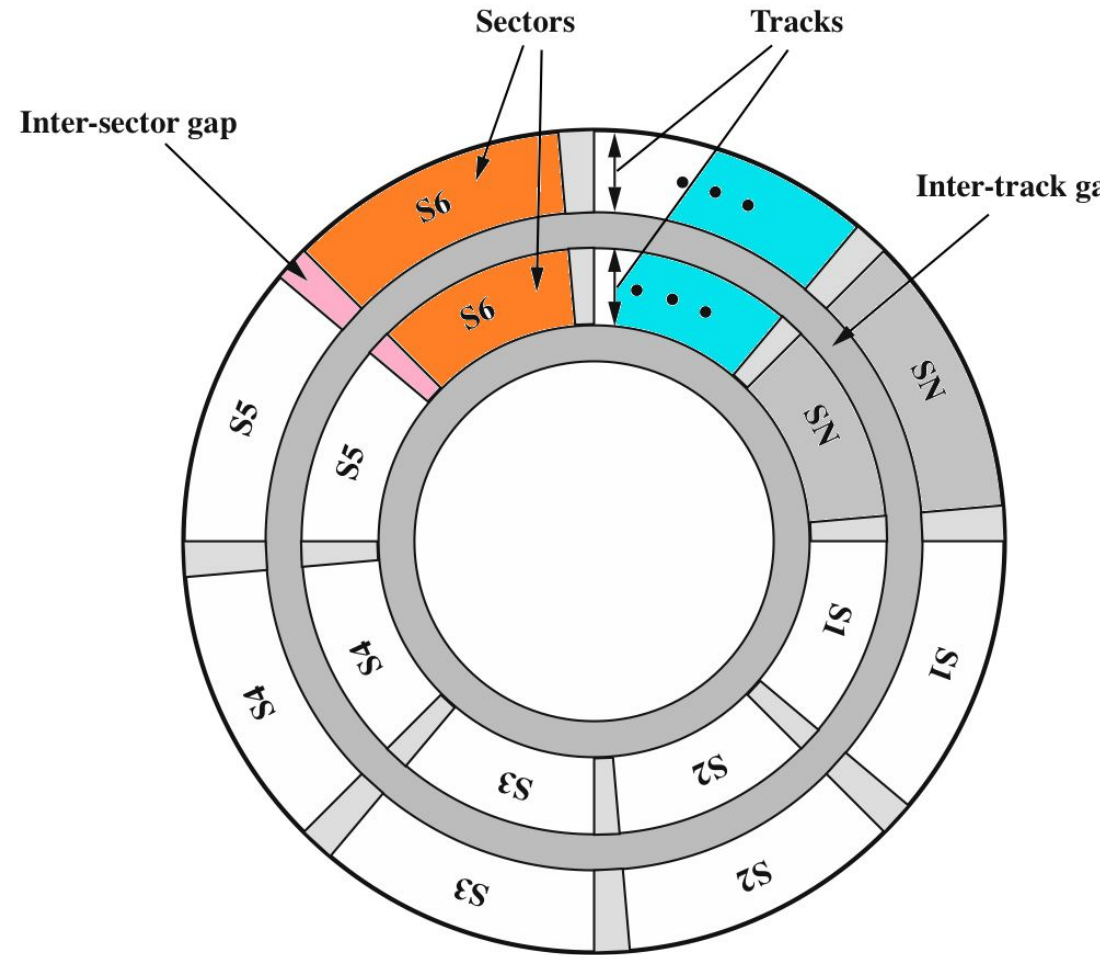
CDs / DVDs



Memory Cards

Data Organization and Formatting

- The disk surface is divided into **concentric rings** called **tracks**.
- Each track matches the **width of the read/write head**.
- Intertrack gaps** separate tracks to reduce errors and interference.
- Each track is further divided into **sectors**, the basic units of data transfer.
- Fixed-length sectors** are commonly used in modern systems.
- Intersector gaps** separate sectors to allow accurate reading and writing.



Improving Secondary Storage Performance



Processor and memory speeds have improved much faster than **disk speeds**.



This **performance gap** made disk storage a **major bottleneck** in computer systems.



To overcome this, designers use **multiple disks working in parallel**.



These **disk arrays** allow:

Parallel handling of multiple I/O requests.
Faster access when data is **distributed across disks**.



This concept forms the basis of **RAID (Redundant Array of Independent Disks)** systems.

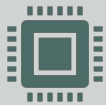
RAID



Using **multiple disks** allows different ways to **organize data** and add **redundancy** for reliability.



However, this variation made it hard to design **universal database systems**.



To solve this, the industry developed a **standardized approach** called **RAID — Redundant Array of Independent Disks**.



RAID provides **better performance, fault tolerance, and data reliability** across platforms.

RAID Features



Appears as **one logical drive** to the operating system.



Data is split across multiple disks using **striping** for faster access.



Extra space stores **parity information** to recover data if a disk fails.

RAID Striping (Data Distribution Technique)

RAID *striping* divides data into equal-sized blocks (called stripes) and stores them across multiple disks.

Data is broken into chunks.

Each chunk is written to a separate disk.

Multiple disks read/write at the same time, improving speed.

Increases performance through parallel access.

No fault tolerance — data loss if one disk fails.

Commonly used in RAID 0, RAID 5, and RAID 10.

Parity



Used to provide **fault tolerance**.



A **parity block** contains information computed from the data blocks (using XOR).



If one disk fails, the missing data can be **reconstructed** from parity and remaining disks.



Common in **RAID 5 and RAID 6** configurations.

RAID: Striping and Parity

Striping:

- Data is divided into equal-sized blocks (called *stripes*) and spread across multiple disks.
- Improves **speed** by allowing simultaneous read/write operations.
- Example: Block 1 on Disk A, Block 2 on Disk B, Block 3 on Disk C...

Parity:

- Adds **redundancy** for fault tolerance.
- A **parity block** stores error-checking information calculated from data blocks.
- If one disk fails, lost data can be **reconstructed** using parity information.

Key Idea:

- Striping = *performance boost*
- Parity = *data protection*
- Together, they balance **speed and reliability** (used in RAID 5 & RAID 6).

Main RAID Levels (0–5)

RAID 0 – Striping

- Data divided across multiple disks (no redundancy).
-  High speed,  no fault tolerance.

RAID 1 – Mirroring

- Data duplicated on two disks.
-  High reliability,  doubles storage cost.

RAID 2 – Bit-level Striping with Error Correction

- Uses extra disks for error-correcting codes (rarely used today).

RAID 3 – Byte-level Striping with Parity

- One disk stores parity; other disks store data bytes.
-  Good for large sequential data transfers.

RAID 4 – Block-level Striping with Parity

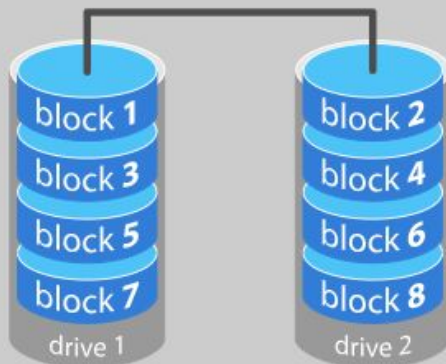
- Parity stored on one dedicated disk.
-  Parity disk can become a bottleneck.

RAID 5 – Block-level Striping with Distributed Parity

- Parity information spread across all disks.
-  Balanced performance, storage efficiency, and reliability.

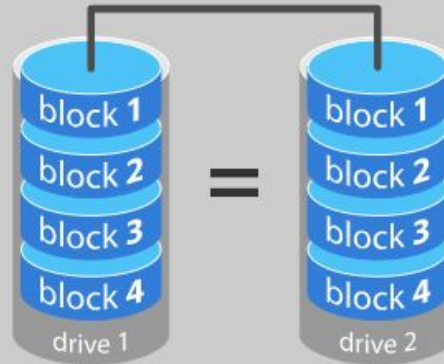
RAID Levels

RAID 0
striping



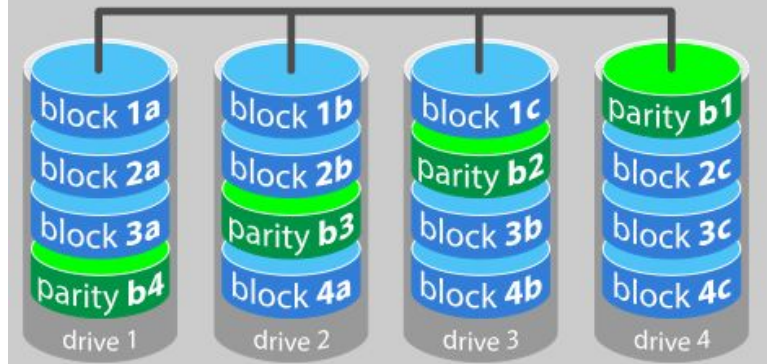
prepressure.com

RAID 1
mirroring



prepressure.com

RAID 5
striping with parity across drives



prepressure.com

Solid State Drives



SSDs are a major modern development in computer storage.



They are used to **replace or support traditional Hard Disk Drives (HDDs)**.



“Solid state” means they use **electronic semiconductor circuits** (no moving parts).



SSDs store data using **NAND flash memory**.



They are **faster, quieter, and more reliable** than HDDs.

SSD vs HDD

	NAND Flash Drives	Seagate Laptop Internal HDD
File copy/write speed	200–550 Mbps	50–120 Mbps
Power draw/battery life	Less power draw, averages 2–3 watts, resulting in 30+ minute battery boost	More power draw, averages 6–7 watts and therefore uses more battery
Storage capacity	Typically not larger than 1 TB for notebook size drives; 4 max for desktops	Typically around 500 GB and 2 TB max for notebook size drives; 10 TB max for desktops
Cost	Approx. \$0.20 per GB for a 1-TB drive	Approx. \$0.03 per GB for a 4-TB drive

Advantages of SSDs over HDDs

Faster performance (high IOPS) – handles more read/write operations quickly.

More durable – no moving parts, so less damage from shocks or vibration.

Longer lifespan – no mechanical wear.

Uses less power – energy-efficient and battery-friendly.

Quiet and cool – no spinning disks or noise.

Much lower access time – over **10× faster** than HDDs.

SSD Organization



The **operating system** uses the **file system** to access stored data.



The **file system** communicates through an **I/O driver** to reach the SSD.



The **I/O driver** manages data transfer between the computer and SSD.



The **interface** connects the SSD to the computer —



e.g., **PCIe** (for internal SSDs) or **USB** (for external SSDs).

Main Components of an SSD

Controller: Manages data flow and runs SSD firmware.

Addressing logic: Selects the correct flash memory cells for reading/writing.

Data buffer / cache: Uses high-speed RAM to boost data transfer speed.

Error correction unit: Detects and fixes data errors automatically.

Flash memory chips: Store the actual data using NAND technology.

Practical Issue 1: Performance Slowdown

SSDs can **slow down over time** as they fill up.

Data is stored in **pages** (≈ 4 KB) inside **blocks** (≈ 512 KB) — only whole blocks can be erased.

Writing new data may require:

- Reading the whole block into memory.
- Erasing the block.
- Writing it back with changes.

As files become **fragmented (scattered)**, more blocks are affected → **slower write speed**.

Solutions



Overprovisioning: Reserve extra space for smooth writes.



Defragmentation: Rearranging the data in idle time to Reduce scattering.



TRIM command: Tells SSD which data blocks are no longer in need that can be erased early.

Practical Issue 2: Limited Write Lifespan



Each flash memory cell can only be **written a limited number of times** ($\approx 100,000$ writes).



Repeated writes cause cells to **wear out** and lose data retention ability.

Solution

Caching: Combine small writes to reduce stress.

Wear-leveling: Spread writes evenly across all cells.

Bad-block management: Avoid damaged areas.

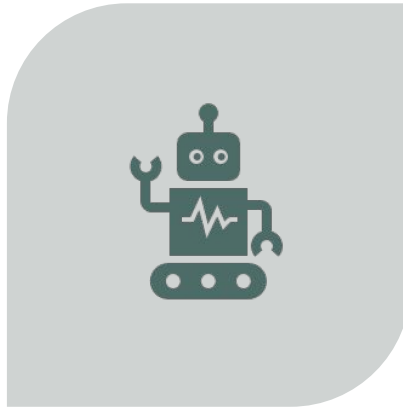
Use of RAID: Improves reliability and reduces data loss risk.

Self-monitoring: SSDs estimate their remaining lifetime.

Modern Storage in Computing: Applications & Practical Use



MODERN SYSTEMS INCREASINGLY
USE **SSD AND FLASH-BASED
STORAGE.**



STORAGE IS CRITICAL IN **IOT,
EMBEDDED SYSTEMS, SBCS, AND
ROBOTICS.**



KEY FEATURES: **FAST ACCESS,
LOW POWER, DURABILITY, AND
RELIABILITY.**

Modern Storage Types



NAND Flash Memory – base technology for most SSDs and embedded storage.



eMMC / UFS – embedded storage in IoT devices, smartphones, SBCs.



NVMe SSDs – high-speed PCIe storage for laptops, servers, and AI systems.

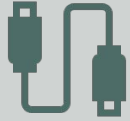


microSD & SD Cards – widely used in SBCs, drones, and portable devices.



USB Flash & Portable SSDs – flexible, high-speed storage options.

New Terms



eMMC (Embedded MultiMediaCard): A type of flash storage commonly used in smartphones and low-cost devices, integrated directly on the motherboard.



UFS (Universal Flash Storage): A faster, more efficient storage standard than eMMC, supporting parallel read/write operations.



NVMe (Non-Volatile Memory Express): A high-speed storage protocol designed for SSDs connected via PCIe, offering extremely low latency and high performance.

Key Applications



IoT Devices: Store sensor data locally before sending to the cloud.



Embedded Systems: Store firmware, configuration, and logs.



Single Board Computers (SBCs): Raspberry Pi, Jetson Nano boot drives and storage.



Robotics & Drones: Store navigation maps, sensor logs, and AI models locally.



Data Centers & Cloud Servers: SSDs for high-speed databases and virtual machines.